DATA FLOW DIAGRAM

Project Name: EduGenie

Technology: Google Gemini Generative AI

Project Type: AI-Powered Learning Assistant

1. INTRODUCTION

The Data Flow Diagram (DFD) represents how data moves through the EduGenie system. It shows the interaction between the student, EduGenie application, Google Gemini AI service and the generated learning response.

2. LEVEL 0 – CONTEXT DIAGRAM

The main external entity is the STUDENT.

STUDENT

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| Educational Content / User Input

↓

EDUGENIE

|

| Processed Request

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GOOGLE GEMINI AI

|

| AI Generated Response

↓

EDUGENIE

|

| Summary / Learning Response

↓

STUDENT

3. LEVEL 1 – DATA FLOW

STEP 1: USER INPUT

The student provides educational content or a learning request through the EduGenie application.

Input:

• Educational text

• Study material

• Learning-related request

STEP 2: INPUT PROCESSING

EduGenie receives the student's input and prepares it for processing.

Process:

• Receive user input.

• Validate the input.

• Prepare the request for the AI model.

STEP 3: AI REQUEST

EduGenie sends the processed request to Google Gemini through the configured AI service/API.

Data Sent:

• User's educational content.

• Required instruction or prompt.

STEP 4: GEMINI AI PROCESSING

Google Gemini processes the request using Generative AI capabilities.

Processing includes:

• Understanding the provided content.

• Identifying relevant information.

• Generating an appropriate response.

• Creating a concise and understandable output.

STEP 5: AI RESPONSE

Google Gemini sends the generated response back to EduGenie.

Output:

• Generated summary.

• Explanation or learning response.

• Relevant educational information.

STEP 6: RESPONSE DISPLAY

EduGenie receives the AI response and presents it to the student through the application interface.

The student can then read and use the generated information for learning and revision.

4. DATA FLOW SUMMARY

Student

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Educational Content / Request

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EduGenie Application

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Input Processing

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Google Gemini AI

↓

AI Processing

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Generated Response

↓

EduGenie Application

↓

Student

5. MAIN COMPONENTS

1. STUDENT

The user who provides educational content and receives the AI-generated learning response.

2. EDUGENIE APPLICATION

The main application that accepts user input, communicates with the AI service and displays the generated response.

3. GOOGLE GEMINI AI

The Generative AI service responsible for processing the educational content and generating the response.

4. USER INTERFACE

The interface through which the student provides input and views the generated output.

6. INPUT DATA

The main input data includes:

- Educational text.
- Study material.
- User questions or instructions.
- Learning-related prompts.

7. OUTPUT DATA

The main output data includes:

- Concise summaries.
- Explanations.
- Important information.
- AI-generated learning responses.

8. CONCLUSION

The Data Flow Diagram shows that EduGenie acts as an intermediary between the student and Google Gemini AI. The student provides educational content, EduGenie processes and forwards the request to Gemini, and the generated response is returned to EduGenie and displayed to the student.